

BT302 FINAL TERM

MCQS+ S.QS+L.QS

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LECTURE FROM 99 TO 167

ALL ABOUT BIOTECH

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BT302 Final Term MCQS

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Topic 99 + 160:

- 1- Hypersensitivity also known as Intolerance
- 2- Abnormal response against the otherwise harmless environment stimulus:
called Allergy
- 3- Allergy causing agent:
food, pollen, animal dander
- 4- Autoimmune disorder:
Abnormal response against self tissues
- 5- Hypersensitivity based on:
Two factors
- 6- How many types of hypersensitivity
Four types (Type I, II, III, IV)
- 7- Type I hypersensitivity also known as
Immediate or Anaphylactic reactions.
- 8- Reaction of Type I hypersensitivity:
Usually 15-30 min, sometimes 10-12 hours.
- 9- Involve various tissues:
Skin, Eyes, Nose, Lungs, GIT

- Imp MCQ
- 10- Type I Hypersensitivity is mediated by:
IgE
 - 11- Which primary cells are involved in Type I:
Basophils, Mast Cells
 - 12- Sensitisation of mast cells with IgE
via which receptor:
FcR receptors
 - 13- EEF-A (tetrapeptides)
It attracts eosinophils and neutrophils
 - 14- Diagnosis of Type I by
By skin + Measurements of total IgE levels
 - 15- Type II Hypersensitivity also known as:
cytotoxic hypersensitivity
 - 16- Type II is specific:
Tissue specific
 - 17- Example of exogenous antigen:
Hemolytic anemia, granulocytopenia and
thrombocytopenia
 - 18- Reaction time of Type II
Minutes to hours
 - 19- Example of Endogenous antigen:
Erythroblastosis Fetalis

- 20- Erythroblastosis Fetalis also called.
Hemolytic disease of newborn (HDN)
- 21- Antibodies against glomerulus
basement membrane: Good pasture's Syndrome
- 22- Antibodies against thyroid:
Hashimoto's thyroiditis
- 23- Type III Hypersensitivity also known as:
Complex hypersensitivity
- 24- Time required for Type III
3-10 hours
- 25- SLE known as:
Systemic Lupus erythematosus
- 26- Specific for various organs:
Skin, Kidney, Lungs, Blood Vessels, Joints
- 27- Antibodies involved in Type III:
Mostly IgG and IgM
- 28- Type 4 Hypersensitivity also known as:
Cell mediated hypersensitivity.
Time Required for Type 4
- 29- 48- 72 hours
- 30- CD8 are
Cytotoxic T cells causes direct damage to
target cells.

31- CD4

Helper T-cells

32- What are Cytokines?

Diverse group of non-antibody proteins that act as mediators between cells.

33- Monokines: produced by mononuclear phagocyte cells.

34- Lymphokines: Produced by activated lymphocytes

35: Interleukins:

They are mediators between leucocytes

36: Chemokine:

They are responsible for leucocytes migration

37: Cytokines function as **cascade** signaling in the form of cytokine network.

38: mRNA

For cytokines are short lived.

39: What is Redundancy?

It is due to the nature of cytokine receptors.

40: Different cells can respond to same cytokine due to:

Structural similarities

41: Cytokine response in how many ways:

Three ways

42: what is Autocrine :

Effect on same cell which secrete cytokine

43: Paracrine:

Effect on nearby cells to cells which secrete cytokine

44: Endocrine:

Effect on distant cells through circulation

45: Cytokines categorized on which basis:

Two basis

46: On basis of source cell what are Monokines?

It produced by mononuclear phagocytic cells e.g. Interferon gamma & Interleukin (IL- 1) from macrophages & Monocytes.

47: On basis of source cell what Lymphokines?

Produced by activated lymphocytes e.g. IL-3, IL-4 & IL-5

48: On basis of source cell what Interleukins?

They are mediators between leucocytes e.g. IL-1, IL-10& IL-18

49: On basis of source cell what Chemokine's?

They are responsible for leucocytes migration e.g. IL-8

50: Types of cytokine on basis of function:

1: Cytokines which act **for innate (non-Specific) immune system** e.g. TNF- α , IL-1, Il 10 etc.

2: Cytokines which act **for Adaptive (Specific) immune system** e.g. Il- 2,IL-4 & IL-5

51: What are Tumor Necrosis Factor-Alpha (TNF- α) :

It is produced by activated macrophages in response to microbes or their products for example lipopolysaccharides (LPS) of gram negative bacteria. It acts a mediator of acute inflammation. It recruits neutrophils and macrophages at the site of infection either by stimulating endothelial cells to secrete adhesion proteins or by secreting chemokines. Moreover, TNF-a induces fever and acute phase proteins in host.

52) IL-1

This is also an inflammatory cytokine as like TNF-a. It is produced by activated macrophages. Furthermore, it also help to activate T-cells.

53) IL-10

It is an inhibitory cytokine which inhibits the cytokine production of cytokines from macrophages.

54) IL-12

This cytokine is produced by dendritic cells and macrophages. It enhances the cytolytic activity of cytotoxic T-cells

55) Type I interferon

This class of interferon include Interferon α & β . These inhibit viral replication in cells by increasing the expression of class I MHC which in turn make them susceptible to cytotoxic T-lymphocytes (CTL).

56) IFN- γ

It is produced primarily by Th1 cells. It enhances the cytolytic activity of NK cells by induction of class I & II MHC molecules.

57) Chemokines

These are chemotactic cytokines produced by many leucocytes. These recruits inflammatory cells at site of infection for an inflammatory response.

58) Which cytokines which act primarily for Adaptive (Specific) immune system?

1) IL-2

It is produced by T-helper (Th) cells but Cytotoxic T-cells (Tc) also produced it in lesser extent. It stimulate B-cell division. It can also activate NK cells and monocytes. Moreover, IL-2 acts on T-cells in autocrine manner.

2) IL-4

IL-4 is produced by macrophages and Th2 cells. It promotes the development of Th2 from naïve T-helper (Th) cells. The Differentiated Th2 cells result in the production of antibodies. IL-4 also involves in the class switching of antibody into IgE

3) IL-5

IL-5 is produced by Th2 cells. It promotes the development of B-cells and eosinophils. It also activates mature eosinophils.

4) Tumor Growth Factor (TGF)- β

TGF- β is produced by T cells and many other cells types. It acts as inhibitory cytokine which inhibits the proliferation of T-cells and activation of macrophages. It also acts on neutrophils and endothelial cells to block the effects of pro-inflammatory cytokines.

59) What are three phases of Immuno-regulation ?

1) Recognition phase

2) Activation Phase

3) Effector Phase

60) What is Immune Evasion?

All the strategies used by pathogenic organisms and tumor cells to evade host immune's system are called as immune evasion mechanisms.

61) What is mechanism of immune evasion?

The strategy to escape from immune system is caused by different microbes differently as follows

a) Extracellular organisms

These organisms usually inactivate the humoral components of immune system

b) Intracellular organisms

These organisms usually bypass the host immune system by inactivating intracellular killing and other cell mediated immune mechanisms

62) What is antigenic variation?

Antigenic Variation

Microorganisms mutate their antigenic surface molecules for immune-evasion which is called as antigenic variation

63) What is antigenic Shift?

Antigenic shift

This is the major form of change in antigenic structure of antigen as a result the new strains of microbes are generated

64) What is antigenic drift?

Antigenic drift

This is the minor form of antigenic variation within the same strain for example influenza (Flu) virus exhibits both antigenic shift and drift for pandemic and seasonal flu respectively. Similarly, bacteria like *E.coli*, *Neisseria gonorrhoeae* and *Salmonella typhimurium* also exhibit antigenic drift.

65) What are three ways of Immune Response against Extracellular Pathogens?

There are three ways of controlling

1) Neutralization

2) Opsonization

3) Complement activation

66)What is Neutralization?

The pathogenicity of bacteria is being determined by secreted molecules of bacteria like toxins. These toxins are neutralized by specific antibodies against toxins in a form of antitoxins by the process of neutralization. The binding of

antibodies with toxins leads towards immune complex formation. Furthermore, the clearance of immune complex is done by the process of phagocytosis.

67) What is Opsonization?

The enhancement of the process of phagocytosis by phagocytes is called as opsonization. The antibodies which bind specifically with extracellular bacteria e.g IgG or IgM are called as opsonins. This result in the formation of immune complex. Moreover, the binding of immune complex occur with specific receptors against Fc fragment of antibodies.

68) What is Complement activation?

The inactive complement proteins are activated by extracellular bacteria in combination with specific antibody. Further, the complement proteins are fixed on antibody which is already bounded with bacterial antigen on bacterial surface. This cause the lysis of bacteria which is the primary function of complement proteins by forming membrane attack complex (MAC).

69) What are two ways of Immune Response against intracellular Pathogens?

Cell mediated immune response varies according to the residing site of the pathogen

1) Cytosolic site 2) Vesicular site

70) What are Cytosolic Pathogens?

The exogenous pathogens are the pathogens which used to reside in the cytosol of infected cell. Viruses control through such mechanism.

71) What are Vesicular Pathogens?

The endogenous pathogens are those pathogens which used to reside in the vesicle like phagosomes of infected cell. Intracellular bacteria e.g *Mycobacterium*, *Listeria*

Mechanisms of Immune Evasion by intercellular Bacteria

Mechanism of Immune Evasion	Examples
Inhibition of phagolysosome formation	<i>Mycobacterium tuberculosis</i> , <i>Legionella pneumophila</i>
Inactivation of reactive oxygen and nitrogen species	<i>Mycobacterium leprae</i> (phenolic glycolipid)
Disruption of phagosome membrane, escape into cytoplasm	<i>Listeria monocytogenes</i> (hemolysin protein)

9

72)

73) What is Cell Mediated Immunity ?

The immune response which is independent of antibodies and based on cell to cell interaction is called as cell mediated immune response or immunity.

74) What is humoral immunity?

Humoral immunity is based on cell free body fluid like serum which contain various protective molecules like antibodies.

75) How Cellular mediated immunity provides protection?

- T-cell immunity

In this form of protection, cytotoxic T-cells (CTL) are activated for the killing of infected and transformed cells

- By activating macrophages

By this way, macrophages are activated for destroying various pathogens

- By activating natural killer (NK) cells

In this way of protection, NK cells are activated for killing of transformed and viral infected cells.

76) Other name of cytotoxic T-Lymphocytes?

Cytotoxic T-Lymphocytes are also called as CTLs and CD8+ve.

77) Functions of cytotoxic T-Lymphocytes

These cells are not mature after exiting thymus where these have functional TCR which can recognize antigens but cannot kill the target cells.

78) How CTL differentiates from Pre-CTLs

In following two signals

1. The specific antigen in context of Class I MHC
2. The cytokines produced from T-helper 1 (Th1) cells especially IL-2 & INF- γ .

79) Functions of CTL: CTLs cause the killing of target cells. This of killing is antigen specific where target cell must bear same antigen with class I MHC molecule

Moreover, CTL killing also requires cell to cell contact in which target cell should contain cell surface MHC molecule. However, CTLs are not injured while targeting the target cell. One CTL can kill sequentially numerous target cells.

78) What is granule-mediated killing?

In this form of killing, fully differentiated CTL have numerous granules which contain perforin & graenzymes in its cytoplasm. Upon CTL degranulation, perforin monomers are polymerized to produce polyperforin molecules. These polyperforin molecules are assembled in the form of channels in the presence of calcium ions.

79) What is role of Role of Helper T-Lymphocytes in Cell-Mediated Immunity?

Helper T (Th)-Lymphocytes which are also called as CD4+ve cells. These cells are at the center of cell mediated immunity. Helper T-lymphocytes recognize specific antigens presented with class II MHC. Moreover, these cells are also involved in B-lymphocytes activation. Helper T-cells also release various cytokines which further activate other immune cells.

80) What are major subtypes of Th cells:

Th1 cells

Th2 cells

81) Briefly describe Th1?

These cells release cytokine like IFN γ upon activation which then activate macrophages. In turn, macrophages participate in the generation & differentiation of CTL for cell mediated immunity. However, IFN γ acts as inhibitory cytokine for Th2.

82) What are Th2 cells?

Th2 cells secrete cytokines like IL-4 & IL-5 which further activate B-cells for humoral immunity. This kind of humoral response is also involved in the generation of IgE antibodies through the process of class switching. Moreover, Th2

cells are also involved in the activation of mast cells and eosinophils for providing immunity against allergen and parasitic infections. Th2 cell also secretes IL-10 which is inhibitory for IFN γ production from Th1 cells.

83) Write Role of Natural Killer (NK) cells in Cell Mediated Immunity

NK cells are also called as **Large Granular Lymphocytes (LGL)** as these resemble with lymphocytes except that they are larger and have numerous granules. These have surface molecules like C16 and CD56 but lack in CD3 molecule. NK cells have the ability of killing virus infected and malignant cells after activation. These are activated upon exposure to cytokines like IL-2 & IFN γ and then recognize the infected and malignant cells. The recognition is based on the absence of class I MHC molecule on the surface of infected cells in contrast to normal cells. Afterward, NK cells kill the target cell like CTL using perforin & granzymes.

84) What is ADCC?

Antibody dependent Cellular Cytotoxicity or (ADCC) is a kind of cellular killing which is based on antibody and killer (K) cells. K cells are not morphologically distinct type of cells. These are any cell which can mediate ADCC like NK cell, macrophages & PMNLs.

85) What is Role of Lymphokine Activated Killer Cells in Cell Mediated Immunity?

Lymphokine activated killer (LAK) cell are NK cells which are differentiated by continuous exposure to IL-2 and interferon. Upon activation, LAK cells kill the transformed and malignant cells. This form killing process is selective for sparing normal cells but need cell to cell interaction.

86) What is Inflammation?

It is the body's attempt in order to protect from harmful stimulus. It is one of the complex tissue responses which is in fact the part of host immune system.

Inflammation can be beneficial for host in order to remold or restore the damaged tissue.

87) What are signs of inflammatory response?

1. Dolor (Pain)

2. Calor (Heat)
3. Rubor (Redness)
4. Tumor (Swelling)
5. Functio laesa (Loss of function)

88) What are Outcomes of Inflammation?

In response of inflammatory process, following outcomes occur

- ☐ There is restoration of normal tissue
- ☐ Large amount of tissue is destroyed which is called as fibrosis
- ☐ There is also pus formation which is also termed as abscess formation
- ☐ Inflammation can be prolonged for longer time points which is also called as chronic inflammation

89) What is Infection?

Infections are not synonym to inflammation. It is the invasion of host tissue by pathogen which is the disease causing microbes for example bacteria, viruses, parasites, and fungi etc.

90) Write Koch's postulates?

I. Infectious agent is present in patients suffering from infection however, absent in healthy individuals II. Infectious agent can grow as pure culture

91) Define Following?

a) Epidemic: this is the sudden outbreak of infection

b) Endemic: this is the constant occurrence of a particular infection in a community or population

c) Pandemic: it is the global occurrence of an infection

92) describe infection according to the action of body after invasion in the form of inflammatory responses?

Infectious diseases can be various kinds according to anatomical locations like

- Respiratory tract infections
- Gastrointestinal (GIT) infections
- Skin infections

- Genital tract infections

92) Role of Phagocytes in Inflammation

The phagocytes are inflammatory cells in nature like macrophages and neutrophils. They have the ability of recognition of infectious agent via receptors on them which are termed as pattern recognition receptors (PRRs). Followings are the different kinds of PRRs

- Toll like receptors (TLRs)
- Fc receptors
- Complement receptors
- Scavenger receptors

93) What are (PAMPs).

PRRs recognize various molecular patterns on the surface of pathogens which are called as pattern associated molecular patterns (PAMPs).

These are in the following forms:

Lipopolysaccharides (LPS) recognized by TLRs

Flagellins: also recognized by TLRs

Cell wall component: through complement receptor after binding with complement.

94) Chemotaxis of Leucocytes in Inflammation

This is the movement of circulating phagocytes towards the site of infection by various chemicals also called as chemotactic agents for example save our soul (SOS) signals from bacteria like N-formylmethionine containing peptides. The clotting system peptides and complement proteins also serve as chemotactic agents. For macrophages, cytokines are considered as good chemotactic agents.

95) Diapedesis

This is the process of crossing the endothelial barrier after squeezing & movement extravascularly towards the site of infection. Infected tissues sites cause the attraction of phagocytes called as chemotaxis and the activation of phagocytes

96) Write two ways of intracellular killing?

There are following two ways of intracellular killing

- 1) Oxygen independent
- 2) Oxygen dependent

97) What is Oxygen independent intracellular killing?

In this process of intracellular killing of bacteria, there is no need of oxygen. The granules and vesicles of phagocytes secrete various enzymes which are hydrolytic proteins. Those proteins are bacteriocidal in nature according to their modes of action.

98) What is oxygen dependent?

There is requirement of oxygen for such kind of intracellular killing of bacteria. It is also termed as “Respiratory Burst” as requirement of glucose and oxygen get increased after phagocytosis. As a result, oxygen containing bacteriocidal radicals are produced.

99) Oxygen Dependent types?

1) Oxygen dependent Myeloperoxidase (MPO) dependent

MPO from granules of phagocytes are involved in this process as a result halide ions (OCl^-) are formed as a result of the following reactions. These are bacteriocidal in nature.

2) Oxygen Dependent Myeloperoxidase (MPO) independent

It is myeloperoxidase independent in which there is involvement of Hexose monophosphate shunt.

100) Explain pro-Inflammatory cytokines?

These are involved in up regulation of inflammatory process. These are predominantly produced from activated macrophages and Th cells like $\text{IL-1}\beta$, IL-6 & $\text{TNF-}\alpha$, IL-12 and $\text{INF-}\gamma$. These have role in mediating innate immune response. Furthermore, various inflammatory diseases like atherosclerosis, cancer and depression also occur due to these cytokines.

101) what is Anti-Inflammatory cytokines

These cytokines are involved in down regulation of inflammatory process. These also control pro-inflammatory cytokine response. In this category of cytokines IL-4 , IL-10 , IL-11 and IL-13 are involved. These have pathological role in systemic inflammatory states. A balance between Pro & anti-inflammatory cytokines is required.

102) Inflammosomes?

A multi-protein oligomer which is responsible for activation of inflammatory responses is called as inflammosomes. These promote the maturation and secretion of pro-inflammatory cytokines like $\text{IL-1}\beta$ and IL-18 .

103) Acute vs. Chronic Inflammation?

Acute Inflammation

This form of inflammation is a short term process occurring in response to tissue injury. It appears in minutes to hours.

Chronic Inflammation

This form of inflammation persists for long time. It is a slow process occurring in response to tissue injury. This process lasts for months to even years.

104) define chemotaxis?

The increased movement of leucocytes towards the site of infection is called as chemotaxis.

105) Types of chronic inflammation?

There are two types of chronic inflammation

1) Non-Specific proliferative inflammation

2) Granulomatous inflammation

106) What is Non-Specific Proliferative Inflammation ?

It is characterized by the presence of non-specific granulation tissue. There is also infiltration of mononuclear cells like Lymphocytes, monocytes and plasma cells. In this process, proliferation of fibroblasts also occur which is called as fibrosis.

Moreover, the proliferation of connective tissues and vessels occur for example in case of lung abscess

107) What is Granulomatous Inflammation ?

It is characterized by the presence of distinct nodular tissue called as granuloma which is the aggregation of activated macrophages with their derived cells called epithelioid cells. Infectious granuloma occur due to infections for example tuberculosis and leprosy.

108) Write Adverse Effects of Inflammation?

Inflammation is associated with general flu like symptoms including

- ☐ Fever
- ☐ Chills
- ☐ Fatigue/Lethargy/Loss of energy
- ☐ Headaches
- ☐ Loss of appetite
- ☐ Body pains
- ☐ Muscle stiffness

109) Define vaccination or immunization?

The ways of providing specific protection against many common and damaging pathogens after stimulation of organism's or individual's immune system is called as vaccination or immunization.

110) Types of vaccination?

1) Active Immunization

It is the induction of immunity after exposure of host to antigen. In this case, antigenic exposure is mandatory. Active immunization can occur naturally by exposing to microbe or other antigen when there is no prior exposure before the entry of antigen. Similarly there is also no pre made antibodies in the host's serum. In this form of immunization, immune system develops antibodies against the microbe after its exposure. This process of antibodies generation is considered slow. Moreover a memory response is also generated in which antibodies remained in use for longer time. This form of immunization can be artificial in nature which is achieved by either injection of microbe before natural exposure or treated microbes or toxins.

2) Passive Immunization

It is the induction of active humoral immunity. In this kind of immunization, antibodies are mandatory. Passive immunization can occur naturally by transfer of maternal antibodies to fetus through placenta. However, the artificial induction can be done by injecting gamma globulins from other individuals or animals to host.

111) Difference between Natural Active Immunization & Artificial active immunization?

1 Natural immunization is acquired immunization which is achieved by natural exposure to antigen or acquiring infection by an infectious agent or microorganism e.g measles virus. This is also termed as natural active immunization. This form of natural exposure to antigen stimulate the humoral immunity as a result specific antibodies are generated against exposed microbe. Moreover, there is also activation of cell mediated immunity for intracellular pathogen.

2 Artificial Active Immunization

Artificial immunization is the kind of immunization which is acquired by artificial exposure to antigen or microorganism which is administrated to body in less virulent or attenuated form like heat killed forms of microbes. This artificial exposure with antigen stimulate the humoral immunity in the form antibodies generation. Moreover, artificial activation of cell mediated immunity for intracellular pathogen for example BCG for *Mycobacterium tuberculosis*. Artificial immunization: acquired immunization by artificial exposure. This artificial

exposure to antigen or microbe which is by administration of less virulent (attenuated), heat killed microbes to organism. Artificial exposure to antigen stimulate the humoral immunity for antibodies.

112) Difference between Natural Passive Immunization and artificial passive immunization?

1. **Natural passive immunization:** It can be acquired immunization which can be obtained by isolating gamma globulins (antibodies) from other's individual. This is called as natural passive immunization. Furthermore, the transfer of gamma globulins (IgG) from mother's blood to fetus via placenta. The placental cells contain FcR γ for IgG transfer to fetal circulation. Similarly, the natural transfer of IgA can be occurring through breast milk to new-born baby.
2. **Artificial Passive immunization**
This form of artificial immunization is the acquired immunization which is achieved by administering gamma globulins or antibodies from others individual's or immune animal's serum for example antibodies against tetanus.

113) Note on Artificial Passive Immunization

This form immunization is done by transferring or injecting of specific gamma-globulins from other individual or from other immune animals. Artificial passive immunity is used against various following acute bacterial infections

- ☐ Diphtheria
- ☐ Tetanus

Moreover, artificial passive immunity is also used against various poisoning conditions like

- Insects or sting biting e.g immunization against various strains of Malaria producing Plasmodium
- Reptiles e.g anti-venom administration in case of snake biting
- Food poisoning (Botulism)

Artificial passive immunity is also used as prophylactic (preventive) measure against various viral infections like Influenza, Poliomyelitis, Measles and Rabies. For artificial passive immunization, antibodies which are used usually from human origin which are **also termed as homologous antisera**. Homologous antisera has

potential risk of transmitting HIV and Hepatitis among individuals. However, these are also raised in other species or animals after immunization **called as Heterologous antisera**. Heterologous antibodies provide immediate protection but also a source of allergies like serum sickness and anaphylaxis.

114) Note on Artificial Active Immunization

This form of immunization is achieved by transferring or injecting live, dead or other components of microbes

Followings are the important vaccines which are used as artificial active immunization

1) Live (attenuated) organisms

2) Killed organisms

3) Sub-unit vaccines

1) Live (attenuated) Vaccines

These vaccines are produced after inactivation of organism by heat for the purpose of loss of pathogen's virulence. These are used against various viral infections like Smallpox, measles, mumps, hepatitis A virus Moreover, these are used as live bacterial vaccine against tuberculosis: for example Bacille Calmette-Guerin vaccine (BCG)

2) Killed Vaccines

Such vaccines are produced after killing of organism by heat, chemical treatment or UV irradiations. These are used against various viral infections like Polio, rabies & influenza. Also, most bacterial vaccines are also killed like typhoid fever, cholera, plague and pertussis.

3) Subunit Vaccines

These vaccines consist of various components of microbes like polysaccharides from capsule and surface proteins. The polysaccharides are T-independent antigens while proteins are T-dependent antigens. These vaccines are used for reducing the toxicity for example Pneumococcus vaccines.

115) Write a note on novel vaccines?

There are also various novel ways of designing vaccines in order to reduce in toxicity of vaccines. These have the potential to provoke both humoral & cell-

mediated immunity. These are mainly used in experimentally for research purpose. However, these would be available for clinical use in future

1) DNA vaccines

These are cloned viral peptides which are transfected into host cell. These vaccines can generate both humoral and cell-mediated response like live attenuated vaccines. Similarly, Anti HIV-DNA vaccines are used in experimental stage which have very low efficacy so far.

2) Immunodominant Peptides

These are simple and easy to prepare. Moreover, these peptides are incorporated with MHC to induce both humoral & cell-mediated responses.

3) Anti-Idiotypic antibodies

These antibodies are used against polysaccharide antibodies. These have long lasting immune response with memory.

116) What is Role of Adjuvant in Vaccination?

These are the substances which increase the antigenicity of weak antigens. These can be of two forms

1) Chemicals

2) Biological

117) What are different kinds of adjuvant?

1) Chemical

These are basically aluminum salt or Alum in its structure. This is the only chemical suitable for human use. These are used in DTP (Diphtheria, Tetanus & Pertussis) vaccine.

2) Biological adjuvants

Various bacteria are used as biological adjuvants. These are basically the bacterial products

3) *B. pertussis* used as adjuvant

4) *M. bovis* (BCG and others)

These are used in combination with oil and detergent for better activation of mononuclear phagocytes and cytokine secretion.

118) what are Adverse Effects of Vaccination?

As live microbe is used for vaccination so these have some adverse effects on host. These effects can be severing, rare, mild and moderate in its nature.

- Joint pains (Arthritis): Rubella vaccination
- Can be fetal: in case of Pertussis
- Neurological disorders: Influenza vaccination
- DPT vaccination has local effects like Redness, swelling and pain
- Mild/Moderate effects: fever, drowsiness. Vomiting and anorexia
- Severe effects: persistent crying, fever, convulsions, collapse, acute encephalopathy, permanent neurological deficit due to pertussis components.

119) what is Tolerance?

The specific immunological unresponsiveness or non-reactivity to an antigen resulting from a previous exposure to same antigen is called as immunological tolerance.

120) Immuno-Tolerance can be break in following two ways:

1) Naturally

In case of autoimmune disorders e.g Rheumatoid Arthritis, SLE etc.

2) Artificially

Exposure to immunosuppressive drugs or X-ray irradiation e.g in case of experimental animals for bone marrow transplantation.

121) Write Mechanisms of Tolerance Induction?

There are following two forms of immunological tolerance

1) Central Tolerance

2) Peripheral Tolerance

Central Tolerance

It occurs in primary lymphoid organs e.g. bone marrow & thymus

Peripheral tolerance

It occurs at secondary lymphoid organs e.g. spleen, lymph nodes, tonsils etc.

122) tolerance is induced against tissues and cells in the following ways:

- By injecting hematopoietic stem cells at neonatal stage as during early development stage there is no development of fully matured lymphoid tissues.
- By immunosuppression e.g. lethal irradiation for killing of host's own stem cells in primary lymphoid organs.
- By using immunosuppressive drugs in order to inactivate immune cells of host
- By Chimeras

These are animals with hybrid nature of immune cells with host's own cells and donor's cells. In this case both transplantation of donor's bone marrow and thymus in early age or by immunosuppression which leads towards functional inactivation of host immune cells.

123) what is Autoimmunity

All mechanisms which are responsible for breakdown of self-tolerance against self-antigens are called as autoimmunity or auto immune reactions.

124) Write note on Classification of Autoimmunity?

Autoimmunity is classified based on the type of tissues or organs involved
There are following two categories of autoimmunity

- Organ-specific autoimmunity
- Non-organ Specific autoimmunity

Organ Specific autoimmunity

In this kind of autoimmunity, the immune response is generated against organ associated specific antigens in the form of auto-antibodies. These antibodies are against organ associated antigens which damage the organ. For organ specific autoimmunity, following organs can be target organs with particular auto-immune disease and the antibody involved (Table.18.1)

- Skin
- Thyroid gland
- Muscles

Non-Organ Specific autoimmunity

In this kind of autoimmunity, the immune response is not against organ associated specific antigens. Similarly, the auto-antibodies which are generated in response to immune response are not against organ associated antigens. This kind auto-immunity effect mainly all those tissues which are present throughout the body like skin, joints, soft tissues etc. That's why these disease are also termed as systemic autoimmune diseases.

125) Write a note on diagnosis and treatment of Autoimmune diseases?

The autoimmune diseases are diagnosed based on symptoms and the nature of autoantibodies produced. Autoantibodies can be against two kinds of antigens

1) Self-cell or cell associated antigens

2) Soluble antigens

3) Biochemical test

Autoantibodies against self-cell/Cell associated antigens

These auto-antibodies are detected using tissues section like kidney, skin etc through immunofluorescence technique. These auto-antibodies can produce following two kind of patterns in immunofluorescence based on the kind of auto-immunity

- Linear Pattern (Goodpastuer Syndrome)
- Granular Pattern (Systemic Lupus Erythramatous)

Autoantibodies against Soluble Antigens

Autoantibodies against soluble antigens are detected using patient's serum for auto-antibodies by following techniques

- Agglutination e.g Anti-nuclear antibodies (ANA) for SLE & RA
- Enzyme linked Immunosorbent Assays (ELISA) e.g Rheumatoid factor for Rheumatoid Arthritis
- Radioimmunoassay (RIA) for detection of anti-thyroid antibodies

Biochemical Assays

Following biochemical assays in the serum of autoimmune patients can be performed for their diagnosis of disease

a) For Intrinsic factor (IF) in case of Pernicious anemia

b) Competition for TSH receptors in case of Grave's disease

c) Auto-reactive T-cells can also be detected by flow-cytometry

Treatment of Autoimmune Diseases

Autoimmune diseases are treated with the goal of reducing symptoms or the better control of autoimmune response in order to increase the quality life of patient.

Moreover, treatment strategy is for increasing the ability of immune system to fight against infections. Treatment varies based on specific disease and symptoms for reducing inflammatory process & immune response. Followings are the drugs used for treating auto-immune diseases

- Anti-Inflammatory drug therapy

Corticosteroid is used which particularly inhibit the cellular signaling for production of pro-inflammatory cytokines. Moreover, it relieves from the symptoms of inflammatory process like pain, redness and fever and reduce inflammatory process

- Immunosuppressive drug therapy

Following immune-suppressive drugs can be used for treating auto-immune diseases

i. Cyclosporine

ii. Cyclophosphamide

iii. Azathioprine

These drugs reduce immune response against self-antigens. Also provide relieving from the symptoms of immune response

- Specific approaches

These kind of strategies using specific antibodies against receptor blocking the effects of immune response generated during auto-immunity. Currently, these approaches are used mainly in research. Followings are the antibodies which are used against their corresponding antigens on the surface of immune cells

- Anti-TNF α receptor antibodies

- AntiIL-2 receptor antibodies

- Anti CD4 antibodies

- Anti TCR antibodies

- Anti-Idiotypic antibodies against autoantibodies

126) define donor and recipient?

One person called as donor who donates tissue to other person termed as recipient.

127) Write kinds of tissue transplantation based on the source of donated tissue?

Isograft

This is a form of tissue which is transplanted between individuals of same genetic makeup like identical twins.

Autograft

It may be from the same person tissue to another site within same individual.

Allograft

This form of transplantation which can be from one person to other non-identical person

Xenograft

This kind of transplantation is across the species for example from animal to human.

127) What are transplantation antigens?

The transplantation antigens which are also termed as Major Histocompatibility Complex (MHC) or Human Leucocytes Antigens (HLA). The MHC exist as a complex which is encoded by group of genes on a same chromosome called **as haplotype**.

128) Types of MHC?

Class I MHC

These genes are for Human MHC which are located at chromosome no. 6. These contain following three major loci

1) Locus B

2) Locus C

3) Locus A

Each major locus encode for polypeptide for MHC structure. In which there is a α -chain that contains antigenic determinants.

Class II MHC: The class II MHC is encoded by genes for Human MHC which are located at chromosome 6. Structurally, the Class II MHC complex is also composed of three following major loci.

1) DP

2) DQ

3) DR

Each of these loci code for one alpha & one beta chain. Both of alpha and beta chains associate together to form class II MHC. Likewise to class I, class II antigens are also polymorphic in nature which means these contain many alleles of α -chains. The DR locus contains more than one beta chain genes and possibly four in various combinations. The Class II MHC is expressed on B-lymphocytes and antigen presenting cells (APC).

129) Write kind of tissue rejection?

1) Host vs. Graft Rejection

In this kind of transplant rejection, there are antigens on the surface of graft which are being recognized as foreign by host immune system

2) Graft vs. Host Rejection

In this form of transplant rejection, the lymphoid (immune) tissues in graft recognize host immune system as foreign as a result immune response is mounted.

130) write Immune Mechanisms of Graft Rejection

The reaction of host against allo-antigens of graft which is also called as host vs graft (HVG) rejection. This is considered as a main obstacle in organ transplantation. The involvement of immune mechanisms in graft rejection is based on

a) Time of rejection

b) Nature of allo-antigens of graft

c) Immune status of host

131) short note According to the time of rejection what are types of graft rejection?

1) Hyper acute rejection

2) Accelerated rejection

3) Acute Rejection

4) Chronic Rejection

Hyper-acute Rejection

This kind of transplant rejection has very quick onset for tissue rejection. It occurs within minutes to hours. The high titer of pre-formed antibodies against antigens of graft are responsible for such form of reactions. Antigen/antibody (immune) reaction occurs on the tissue surface. Afterward the fixation of complement leads towards graft destruction

Accelerated Rejection

This kind of rejection is also called as secondary or 2nd set rejection. It occurs after transplantation of second graft. It is due to sharing of antigenic determinants with the first transplant. Time required for this form of rejection is within 2-5 days. The sensitized T-cells during first graft are responsible for this kind of rejection. Moreover, the Lymphokines and cytotoxic T-lymphocytes (CTLs) augment it.

Acute Rejection

It is also called as primary or 1st set of tissue rejection. It occurs during first graft with allo-antigen on grafted tissue. The time span required for such form of rejection is 1-3 weeks. These reactions are mediated by sensitized T-cells to class I & II of allo-graft. Furthermore the secretion of Lymphokines leads towards the activation of monocyte and macrophages.

Chronic Rejection

This form of transplantation rejection is called as delayed rejection which occur within months to years. After transplantation the graft remains normal for months to years but sudden rejection occur due to still unknown mechanisms. These are many hypotheses reading such form of rejection like the involvement of various infections which are responsible for loss of immunological tolerance by grafted tissue in host.

132) prevention and treatment of Graft rejection?

Prevention & Treatment of Graft Rejection

If there is decrease in tissue rejection then there is relative increase in survival of graft. There are successful grafts occur for example mostly in case of kidney & cornea. For successful transplantation there is need of better understanding of immune response and MHC

The success of tissue graft is based on following important factors

1) Donor selection

2) Recipient preparation

3) Immunosuppression

Donor Selection

For appropriate donor selection, there should be MHC compatibility with recipient. Identical twins are considered as the ideal donor which are also termed as Isograft. HLA matched siblings have 95-100% chance of graft success. One haplotype of parent or sibling must be HLA-D matched for successful graft. Moreover, ABO compatibility is also essential for high success rate of transplant

Recipient Preparation

For successful tissue graft, the recipient should be in good health with no current infection. Similarly the absence of active malignancy would also increase the success rate. Moreover, in recipient the absence of any systemic diseases would lead towards better rehabilitation. Likewise, the recipient should not be hypertensive. One to five transfusions of 100-200 ml of donor's blood at 1-2 weeks interval would also increase the compatibility level with donor.

Immunosuppression

This is the way through which host immune system can be suppressed and is most essential component of allo-transplantation. For this there is usage of following immunosuppressive drugs with their specific mode of actions

- Cyclosporin A

It inhibit IL-2 synthesis following antigen exposure to host as a result no immune response is generated for allo-antigens.

- Rapamycin

It inhibits signal transduction for the activation and differentiation of immune cells. For instance inhibition of T-cells proliferation and activation is achieved by this.

133) write two main types of tumor associated antigens ?

1) Tumor specific transplantation antigens

These antigens are unique to tumor cells. These are not expressed on normal cells. These antigens are responsible for rejection of tumor by host immune system. In most cases, these antigens cannot be easily identified

2) Tumor Associated transplantation antigens

These antigens are expressed by both tumor & normal cells in contrast to tumor specific transplantation antigens. Various chemicals, UV radiations & viruses are

responsible for expression of these neo-antigens. Majority of these tumors are weakly immunogenic or on-immunogenic.

134) Define Onco-fetal antigen?

Kind of antigen that are expressed during early development & lost during adult life are called as onco-fetal antigens.

135) Write two Types of onco-fetal antigens?

1) Alpha-fetoproteins (AFP)

2) Carcino-embryonic antigens (CEA)

136) what are Alpha-fetoproteins (AFP)

These antigens are found as secretory protein in serum. Their level get raised during hepatocellular carcinoma.

137) define Carcino-embryonic antigens (CEA)

These antigens are found both in secretory & cell associated form. Their level get raised in colon cancer.

138) NOTE ON Immuno-surveillance mechanisms?

The mechanism which is responsible for tumor rejection in host is termed as Immuno-surveillance mechanisms.

Tumor cells develop strategies for evading immune surveillance. Tumors escape from immune response by following mechanisms

I. Tumor cells may not express neo-antigens which are immunogenic in nature as a result no immune response is generated against such tumors

II. Tumor cells may fail to express co-stimulatory molecules for activation of T-lymphocytes for generating effective cell mediated immune response against tumors

III. Certain tumors may lack or poor expression of MHC for proper immune response generation

IV. At early development of tumor, there is low level of antigens which are insufficient for activation of immune system which is termed as low dose tolerance

V. Overwhelming of immune system, after sudden maturation & expression of neo-antigens on tumor cells which is called as high dose tolerance

VI. Certain tumors secrete immunosuppressive molecules for inactivation of immune cells

VII. Some tumors shed their antigens for neutralization of antibodies

139) Write Use of Tumor Neo-antigens in Immuno-diagnosis?

Tumor neo-antigens are used for *In-vivo* detection of relatively small tumor foci (masses) through radiolabelled monoclonal antibodies against tumor antigens. Likewise the *In-vitro* use for finding the cell origin of undifferentiated tumor particularly lymphocytic origin in the blood & bone-marrow. Furthermore, Immuno-histochemical use of monoclonal antibodies against metastatic foci of tumors can help in the diagnosis of such tumors.

140) Write use of Neo-antigen in immunotherapy?

The tumor neo-antigens can also be used for Immunotherapy for treating various tumors in different following ways

1) Active immunotherapy

In this form of immunotherapy, host actively participates against tumors. Active immunotherapy is achieved by inducing immune response by injecting irradiated tumor cells having neo-antigens on them. The active form of immunotherapy has following two forms

a) Non-specific active immunotherapy

This is generated for activation of immune response against neo-antigens using various vaccines e.g BCG for activation of macrophages against tumors

b) Specific active immunotherapy

In this form, the killed tumor cells or their antigens are used for killing of tumor cells

2) Passive immunotherapy

This form of immune-therapy uses the pre-formed antibodies against specific neo-antigens of tumor through following ways. The specific monoclonal antibodies can be used against tumor antigen can be served for

a) As a vehicle for delivering anti-cancer drug to tumor or cancer

b) Activation of components of innate immune system like complement proteins

141) Define immunodeficiency?

The deficiency or failure of immune system is called as immunodeficiency. It is a state of complete absence of immunity or effective immune response.

142) Explain classification of Immunodeficiency?

Classification of Immunodeficiency

Immunodeficiency is classified into following two main categories based on mode of its acquisition

- 1) Primary Immunodeficiency
- 2) Secondary Immunodeficiency

Primary Immunodeficiency

Primary immunodeficiency is basically the inherited or congenital defects of the immune system. These defects can be either in specific or non-specific immune systems. These are classified on the basis of the site of lesion in the developmental or differential pathway of immune system. The chances of susceptibility to variety of infections get increased in this kind of immunodeficiency. The nature of infection depends on the kind of immunodeficiency.

Primary immunodeficiency can be of two kinds

Immunodeficiency in specific immune system

This kind of immunodeficiency is due to defects in stem cell differentiation. Followings are the important disease of immunodeficiency in specific immune system.

Reticular dysgenesis

It is either complete absence or severe deficiency of lymphocytes & granulocytes in blood due to decrease differentiation of bone marrow stem cells.

Severe combined immunodeficiency (SCID)

This disorder is due to the involvement of lymphoid stem cells. There is complete absence of T & B cell immunity. The patients suffering from SCID are susceptible to variety of bacterial, viral, mycotic & protozoan infections.

DiGeorge Syndrome

This is the disorder of T-lymphocytes due to congenital thymic aplasia or hypoplasia. This condition is highly immunodeficient as live vaccines also cause infections in the patients suffering from DiGeorge syndrome.

X-linked hypo gammaglobulinemia

This kind of immunodeficiency is due to functional disorder of B-Lymphocytes. Such patients are highly susceptible to recurrent bacterial infections.

IgA deficiency

There is deficiency of IgA in this disorder as a result individuals are susceptible to mucosal surface infections like GIT, eye & nasopharynx.

Selective IgG deficiency

In this kind of immunodeficiency, there is selective deficiency of IgG which leads towards hyper IgM immunodeficiency.

Immunodeficiency in Non-Specific immune system

Followings are the important conditions which are linked with immunodeficiency in non-specific immune system

Cyclic Neutropenia

There is low number of circulating neutrophils in the blood which cause recurrent bacterial infections among such patients.

Chronic Granulomatous disease (CGD)

This kind of immunodeficiency is due to defects in phagocyte functions.

Complement deficiency

There is deficiency of complement proteins in this kind of immunodeficiency. Patients suffering from such disorder are susceptible to infections particularly *Neisseria*

Secondary Immunodeficiency

This kind of immunodeficiency is not congenital in nature in contrast to primary immunodeficiency. This state of immunodeficiency is associated with infections for example AIDS. There is also association of aging factor like hypo-cellularity in bone marrow. Malignancies for example Leukemia and Myeloma are also linked with secondary immunodeficiency. Furthermore, other metabolic disorder like diabetes and renal malfunction etc are also linked with secondary immunodeficiency.

143) What is AIDS?

Acquired Immunodeficiency Syndrome (AIDS)

This condition is caused by a virus called as human immunodeficiency virus (HIV). This condition is initiated through an initial infections with influenza like illness. Further, the disease progress with other defects in immune system leading towards severe immunodeficiency. The patients get severe opportunistic infections

like tuberculosis. Moreover, the chance of getting tumor also increased. AIDS is linked with abnormalities in circular lymphocytes.

144) Transmission of AIDS?

For AIDS, HIV infection is transmitted by following two modes

1) Horizontal transmission

This kind of transmission occur usually by person to person contact either sexually or through body fluids like blood and semen

2) Vertical transmission

This form of transmission is from mother to fetus during the process of pregnancy, delivery & breast feeding.

145) define Ataxia-telangiectasia

This form of T-cells associated immunodeficiency has deficient T-cells with less movement of blood vessels which also effects the function of T-lymphocytes.

146) define Wiskott-Aldrich Syndrome

In this condition, there is normal number of T-cells with reduced functions as a result chances of getting multiple infections become increased.